Database Design Report

for

Customer Loyalty Program Database

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NMAD

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Revision History

Version	Date	Description of Changes
CP#1	20.10.2025	Introducing and making data requirement and ER diagram for Customer Loyalty Program Database
CP#2	16.11.2025	Correcting layout of ER model and based of it making optimized Logical Schema
CP#3	08.12.2025	Making a logical schema in Crow's Foot notation in MySQL Workbench and the code in sql
…		

1 Introduction

A customer loyalty program database is needed to attract more customers and make them satisfied by the help of a bonus system. Customers install the application, register, create an account and enter their data: name, surname, telephone number and address. After that they get a loyalty card so they can accumulate points which are then exchanged for discounts, gifts or free goods.
All customer data, their cards, accumulated points, purchases, as well as information about purchases itself, company employees and branches are stored in the database.
The advantage of such a base is that it is convenient for customers, employees, managers - everything is in one place.
Customers can quickly take a look at the news, the amount of their points, history purchase and available awards. For every purchase they are automatically awarded points.
Managers, with the help of collected data, see which goods are being purchased the most, what customers want, how many people participate in the program. This helps them to improve their marketing and to better understand their customers' behavior. Also they can compare the work of other branches, and find out which products and awards are the most popular.
Employees can easily register the new customer, update data of existing customers (e.g. a new phone number) and quickly help with purchases or accrual of points - without paperwork.
Reports in the database show how many new customers have been added, how many points have been awarded and used, which branches work better than others.
One of the biggest advantages - all information and data collected in one place. This reduces the amount of mistakes, saves time, and helps to find the necessary data easily. For instance, who has the most points, which awards are being used the most or which gifts the customers are waiting for. The system can even send notifications about new discounts and available awards, so that the company can keep in touch with customers.
The database also helps to answer important questions:
Who comes most often?
Which awards do people prefer?
Which branch does the most loyalty program transactions?
How many points have been used in a month?
How many awards have the customers got?
Answering these questions, the company can make smarter decisions, improve the service and increase the customer satisfaction.
In general, the database unites everything: from customers to employees, from goods to bonuses. This makes the work clearer and simpler, saves time and strengthens the customer trust, so that they come back again. For companies this is a good way to develop and stand out among competitors.

2 Related systems

The IKEA Family Program, an offer of free loyalty [1] is the initiative focused on offering some value to the regular customers and ensuring the shopping experience becomes better. These benefits are provided at IKEA Family, which has more than 150 million members all over the globe whether online or through the store. IKEA Family advantages: Rewards during a birthday: As a gift a free meal and a 15 dollars in-store coupon. Discounts: Members also receive discounts on specific merchandise and a blanket discount of 5 percent off store goods making a purchase on merchandising goods that qualify. Free coffee: This is the one that you get in the store of IKEA even as you enter it; you will have a cup of free hot drinks that will keep you on the move. Member events: The styling events to members include workshops on home furnishing. Purchase Protection: the program offers Oops-assurance program in which damaged furniture is given complimentary replacement. Buy back & resell: The members will find IKEA Buy Back and Resell program where the members have a choice of reselling products that are used in order to earn store credit.

Starbucks Rewards [2] is one of the most successful loyalty schemes throughout the globe and its membership consisted of over 21 million customers as of 2021. This is what they have been able to offer convenience and value to the customers and build brand following, which can be ascribed to the success of the program. So what is so special about the program? Advantages of Starbucks Rewards: Customer flexibility: Starbucks loyalty program enables customers to do their shopping the way, time and place they choose, integrated in the web, mobile and in store. Exclusive and personally-benefitting members: These include the members getting free refills in-store and reward treats on their birthday, among other benefits that are personalized. Program operation: earn stars through daily shopping then redeem stars for more Starbucks product or you can get merchandise that is not available to everyone.

Nike [3] as a reward provider is significantly above average as the company offers unique specifications, tailor-made experience, and a community orientation through their apps, SNKRS, Nike Run Club and Nike Training Club. Member products: members get the first mover benefits in products unavailable to the other part of the world. Individualized presents: Nike leverages its records on their members to offer them customized recommendations in the form of birthday offers and recommendations. Community: Nike can create a community through the integration of its set of fitness applications where it encourages the community to be part of different challenges, run guidance, and community run. Free shipping where the order is more than 50 dollars and free return receipts.

Korzinka [4] opened its first supermarket in 1996, pioneering modern retail in Uzbekistan. In 2005, it launched korzinka.uz, the country's first online supermarket. Their loyalty program Korzinka Plus, can have unregistered users that still gets 0.5% cashback, and registered users that can get up to 3% cashback. Moreover, for registered users if their monthly purchase amount 1,300,000 soums, the Customer who has registered accordingly receives an additional 1% retro bonus from the amount of the monthly purchases

3 Data Requirements

The data requirements for the proposed loyalty program database suggested to be all relevant information related to customers, transactions, rewards, branches, staff activity, and promotional campaigns in franchise chain-based business. The system store information about customers and their enrollment in the loyalty program, their accumulated points, products offered at different branches, employee assignments, inventory levels, and purchase transactions that earns loyalty points. In addition, the system keeps track of promotional campaigns initiated by the company and their delivery to targeted customers. These data elements help to support the business goals of increasing customer retention, monitoring franchise performance. To sum up it is great tool for promotion management across all branches.

3.1 Conceptual Data Requirements

Before presenting the ER diagram, this subsection helps to understand the relationships between real-world elements to specify the conceptual structure of the loyalty program. In the form of relationships between entity sets, the following assertions delineate the core business rules of the system. These statements provide a high-level description of the system's logical behaviors without going into specifics about its features or implementation. The Entity-Relationship model is built in Section 4 using the conceptual definition that is shown below as the basis.

3.1.1 Entities and Relationships – Specification by sentences

1. A **Customer** registers for the **Loyalty Program**.
2. The **Loyalty Program** has many **Customers**.
3. A **Customer** shops at any **Branch**.
4. A **Customer** makes one or many **Transactions**.
5. Each **Transaction** is associated with exactly one **Customer**.
6. A **Customer** redeems multiple **Rewards**.
7. A **Transaction** includes one or more **Products**.
8. A **Product** appears in many **Transactions**.
9. **Points** are earned through **Transactions**.
10. A **Product** belongs to one **Product Category**.
11. A **Product Category** groups many **Products**.
12. A **Reward** requires a minimum number of **Points**.
13. The **Loyalty Program** offers many **Rewards**.
14. A **Reward** applies to eligible **Customers**.
15. A **Branch** offers many **Products**.
16. A **Branch** employs many **Employees**.
17. An **Employee** works at exactly one **Branch**.
18. A **Branch** maintains an inventory of **Products**.
19. An **Employee** manages many **Transactions**.
20. Each **Transaction** is managed by exactly one **Employee**.
21. An **Employee** registers many **Customers**.
22. The **Company** has one or many **Branches**.
23. The **Company** launches many **Promotions**.
24. A **Promotion** targets a selection of **Customers**.

25. A **Promotion** features specific **Rewards**.
26. A **Reward** can be featured in many **Promotions**.
27. A **Customer** gets notified about **Promotions.**
28. An **Employee** approves many **Promotions**.
29. Each **Promotion** is approved by exactly one **Employee**.

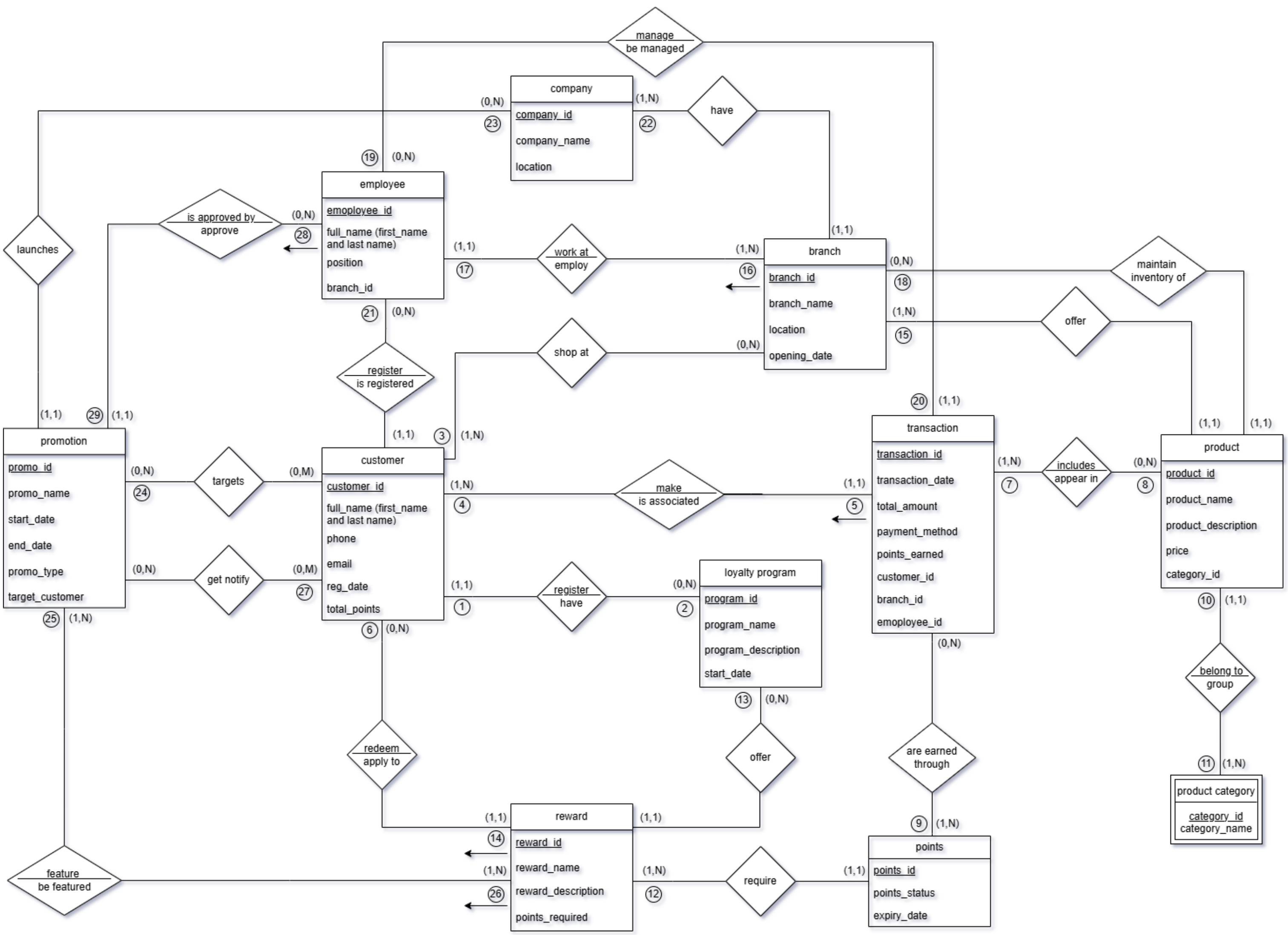
3.1.2 Entities and relationships' attributes

1. Customer (entity)
   Attributes: customer_id (simple), full_name (composite), phone (simple), email (simple), reg_date (simple) and total_points (derived)
2. Loyalty program (entity)
   Attributes: program_id (simple), program_name (simple), program_description (simple) and start_date (simple)
3. Company (entity)
   Attributes: company_id (simple), company_name (simple) and location(composite)
4. Branch (entity)
   Attributes: branch_id (simple), branch_name (simple), location (composite), opening_date (simple) and company_id (simple)
5. Employee (entity)
   Attributes: employee_id (simple), full_name (composite), position (simple) and branch_id (simple)
6. Product (entity)
   Attributes: product_id (simple), product_name (simple), product_description (simple) and category_id (simple)
7. Product category (entity)
   Attributes: category_id (simple) and category_name (simple)
8. Transactions (entity)
   Attributes: transaction_id (simple), transaction_date (simple), total_amount (simple), payment_method (simple), points_earned (simple), customer_id (simple), branch_id (simple) and employee_id (simple)
9. Points (entity)
   Attributes: point_id (simple), point_status (simple) and expiry_date (simple)
10. Reward (entity)
    Attribute: reward_id (simple), reward_name (simple), reward_description (simple) and points_required (simple)
11. Promotion (entity)
    Attributes: promo_id (simple), promo_name (simple), start_date (simple), end_date (simple), promo_type (simple) and target_customer (composite)

4 Conceptual Entity-Relationship Design

We have graphically illustrated conceptual ER diagram using data structure and business rules which are written in Section 3. The diagram shows relationships between loyalty program entities across franchise-based network. Also, relationships between consumers, staff, branches, transactions and etc. are shown in our diagram. We used circled tags to show that each of them represent business-rule sentences written in Section 3.11. Every relationship in diagram helps to understand connections between entities.

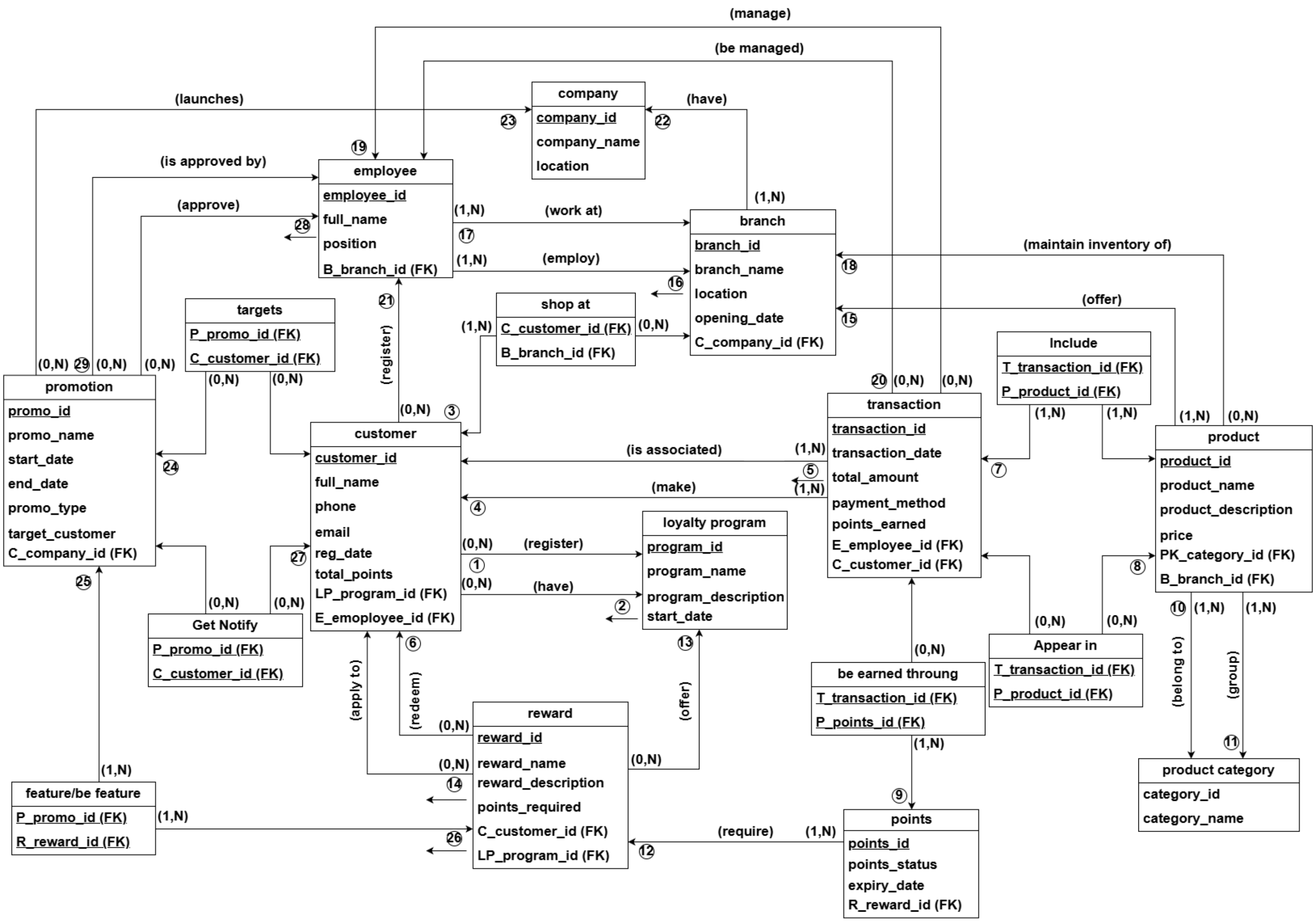
4.1.1 ER/EER-diagram



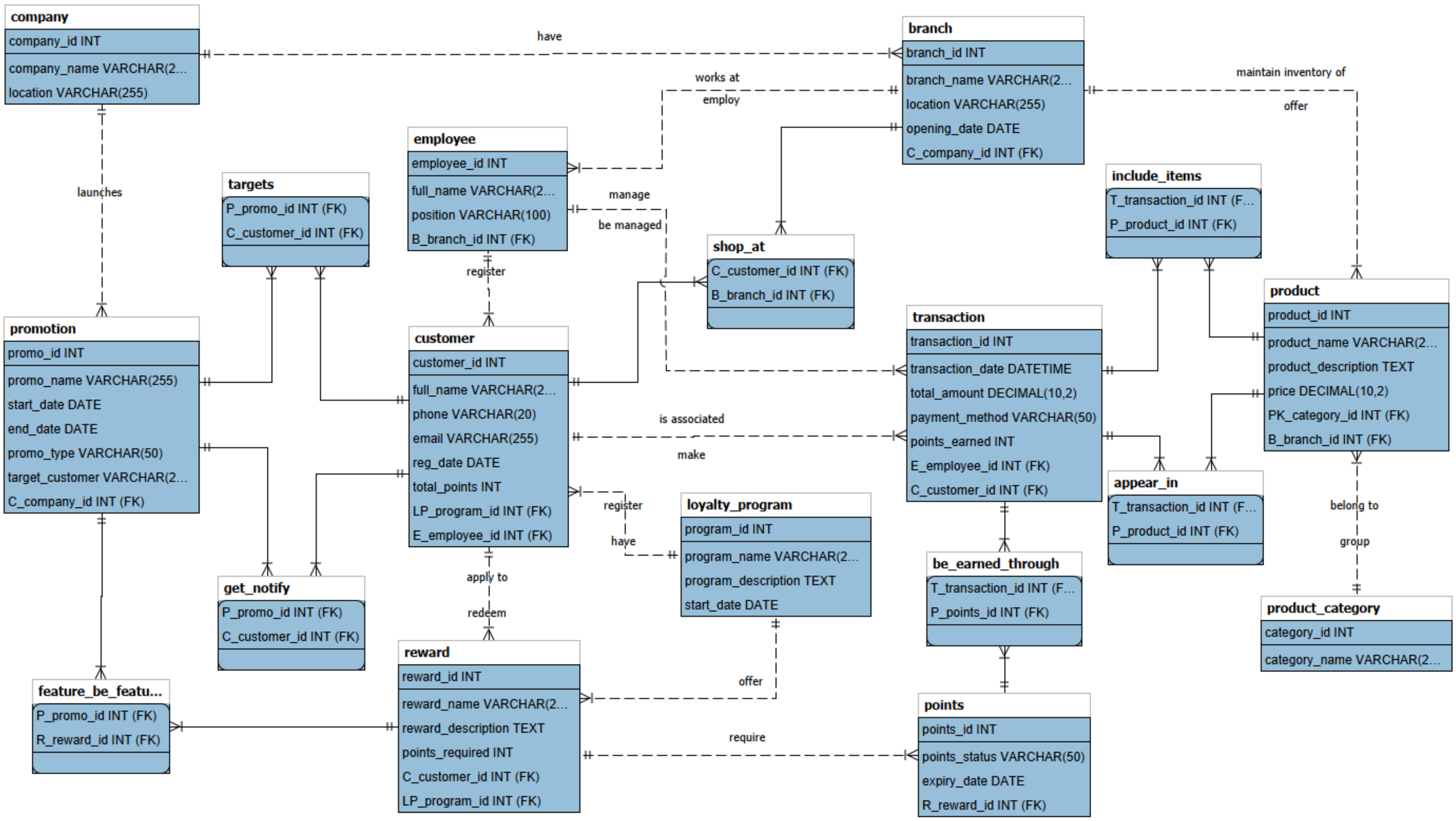
5 Logical Design (*Optimized* Logical Schema)

In this section there is the optimized logical schema that is made from the conceptual ER/ERR diagram in Section 4.1.1. Entities are mapped to relational tables, one-to-many relationships are implemented through foreign keys and many-to-many relationships are turned into tables. In our schema primary keys are underlined and foreign keys are marked as FK.

5.1 Optimized Logical Schema using "Arrows" Notation



5.2 Optimized Logical Schema by "Crow's foot" Notation (MySQLWB)



6 DDL Schemas you created from Logical Schema

```
CREATE TABLE company (
    company_id       INT PRIMARY KEY,
    company_name     VARCHAR(255) NOT NULL,
    location         VARCHAR(255) NULL
);

CREATE TABLE product_category (
    category_id      INT PRIMARY KEY,
    category_name    VARCHAR(255) NOT NULL
);

CREATE TABLE loyalty_program (
    program_id       INT PRIMARY KEY,
    program_name     VARCHAR(255) NOT NULL,
    program_description TEXT NULL,
    start_date       DATE NOT NULL
);

CREATE TABLE branch (
    branch_id        INT PRIMARY KEY,
```

```
    branch_name      VARCHAR(255) NOT NULL,
    location         VARCHAR(255) NOT NULL,
    opening_date     DATE NULL,
    C_company_id     INT NOT NULL,
    FOREIGN KEY (C_company_id) REFERENCES company(company_id)
);

CREATE TABLE promotion (
    promo_id         INT PRIMARY KEY,
    promo_name       VARCHAR(255) NOT NULL,
    start_date       DATE NOT NULL,
    end_date         DATE NOT NULL,
    promo_type       VARCHAR(50) NOT NULL,
    target_customer      VARCHAR(255) NULL,
    C_company_id     INT NOT NULL,
    FOREIGN KEY (C_company_id) REFERENCES company(company_id)
);

CREATE TABLE employee (
    employee_id      INT PRIMARY KEY,
    full_name        VARCHAR(255) NOT NULL,
    position         VARCHAR(100) NOT NULL,
    B_branch_id      INT NOT NULL,
    FOREIGN KEY (B_branch_id) REFERENCES branch(branch_id)
);

CREATE TABLE product (
    product_id       INT PRIMARY KEY,
    product_name     VARCHAR(255) NOT NULL,
    product_description TEXT NULL,
    price        DECIMAL(10, 2) NOT NULL,
    PK_category_id   INT NOT NULL,
    B_branch_id      INT NOT NULL,
    FOREIGN KEY (PK_category_id) REFERENCES
product_category(category_id),
    FOREIGN KEY (B_branch_id) REFERENCES branch(branch_id)
);

CREATE TABLE customer (
    customer_id      INT PRIMARY KEY,
    full_name        VARCHAR(255) NOT NULL,
    phone        VARCHAR(20) NOT NULL,
    email        VARCHAR(255) NULL,
    reg_date         DATE NOT NULL,
    total_points     INT NOT NULL DEFAULT 0,
    LP_program_id    INT NULL,
    E_employee_id    INT NULL,
    FOREIGN KEY (LP_program_id) REFERENCES loyalty_program(program_id),
    FOREIGN KEY (E_employee_id) REFERENCES employee(employee_id)
);

CREATE TABLE reward (
    reward_id        INT PRIMARY KEY,
    reward_name      VARCHAR(255) NOT NULL,
```

```
    reward_description  TEXT NULL,
    points_required     INT NOT NULL,
    C_customer_id   INT NULL,
    LP_program_id   INT NOT NULL,
    FOREIGN KEY (C_customer_id) REFERENCES customer(customer_id),
    FOREIGN KEY (LP_program_id) REFERENCES loyalty_program(program_id)
);

CREATE TABLE points (
    points_id       INT PRIMARY KEY,
    points_status   VARCHAR(50) NOT NULL,
    expiry_date     DATE NOT NULL,
    R_reward_id     INT NULL,
    FOREIGN KEY (R_reward_id) REFERENCES reward(reward_id)
);

CREATE TABLE transaction (
    transaction_id  INT PRIMARY KEY,
    transaction_date    DATETIME NOT NULL,
    total_amount    DECIMAL(10, 2) NOT NULL,
    payment_method  VARCHAR(50) NOT NULL,
    points_earned   INT NOT NULL,
    E_employee_id   INT NOT NULL,
    C_customer_id   INT NOT NULL,
    FOREIGN KEY (E_employee_id) REFERENCES employee(employee_id),
    FOREIGN KEY (C_customer_id) REFERENCES customer(customer_id)
);

CREATE TABLE shop_at (
    C_customer_id   INT,
    B_branch_id     INT,
    PRIMARY KEY (C_customer_id, B_branch_id),
    FOREIGN KEY (C_customer_id) REFERENCES customer(customer_id),
    FOREIGN KEY (B_branch_id) REFERENCES branch(branch_id)
);

CREATE TABLE targets (
    P_promo_id      INT,
    C_customer_id   INT,
    PRIMARY KEY (P_promo_id, C_customer_id),
    FOREIGN KEY (P_promo_id) REFERENCES promotion(promo_id),
    FOREIGN KEY (C_customer_id) REFERENCES customer(customer_id)
);

CREATE TABLE get_notify (
    P_promo_id      INT,
    C_customer_id   INT,
    PRIMARY KEY (P_promo_id, C_customer_id),
    FOREIGN KEY (P_promo_id) REFERENCES promotion(promo_id),
    FOREIGN KEY (C_customer_id) REFERENCES customer(customer_id)
);

CREATE TABLE feature_be_feature (
    P_promo_id      INT,
```

```
    R_reward_id     INT,
    PRIMARY KEY (P_promo_id, R_reward_id),
    FOREIGN KEY (P_promo_id) REFERENCES promotion(promo_id),
    FOREIGN KEY (R_reward_id) REFERENCES reward(reward_id)
);

CREATE TABLE include_items (
    T_transaction_id    INT,
    P_product_id    INT,
    PRIMARY KEY (T_transaction_id, P_product_id),
    FOREIGN KEY (T_transaction_id) REFERENCES
transaction(transaction_id),
    FOREIGN KEY (P_product_id) REFERENCES product(product_id)
);

CREATE TABLE appear_in (
    T_transaction_id    INT,
    P_product_id    INT,
    PRIMARY KEY (T_transaction_id, P_product_id),
    FOREIGN KEY (T_transaction_id) REFERENCES
transaction(transaction_id),
    FOREIGN KEY (P_product_id) REFERENCES product(product_id)
);

CREATE TABLE be_earned_through (
    T_transaction_id    INT,
    P_points_id     INT,
    PRIMARY KEY (T_transaction_id, P_points_id),
    FOREIGN KEY (T_transaction_id) REFERENCES
transaction(transaction_id),
    FOREIGN KEY (P_points_id) REFERENCES points(points_id)
);
```

7 DML Statements used to Insert Data to Schema

```
-- Insert data into company
INSERT INTO company (company_id, company_name, location) VALUES
(1,'Global Pharm','Tashkent'),
(2,'HealthPlus','Samarkand'),
(3,'MediCare','Bukhara'),
(4,'PharmaLife','Fergana'),
(5,'Wellness Co','Andijan'),
(6,'BioPharm','Namangan'),
(7,'LifeCare','Nukus'),
(8,'MedWorld','Termez'),
(9,'PharmaMax','Karshi'),
(10,'HealthFirst','Jizzakh'),
(11,'MediTrust','Kokand'),
(12,'WellCare','Chirchik'),
```

```
(13,'PharmaGen','Navoi'),
(14,'VitaLife','Margilan'),
(15,'HealthyWay','Urganch');

-- Insert data into product_category
INSERT INTO product_category (category_id, category_name) VALUES
(1,'Vitamins'),
(2,'Pain Relief'),
(3,'Cold & Flu'),
(4,'Skin Care'),
(5,'Supplements'),
(6,'First Aid'),
(7,'Digestive Health'),
(8,'Oral Care'),
(9,'Allergy'),
(10,'Weight Loss'),
(11,'Eye Care'),
(12,'Heart Health'),
(13,'Children Care'),
(14,'Herbal'),
(15,'Diabetes Care');

-- Insert data into loyalty_program
INSERT INTO loyalty_program (program_id, program_name,
program_description, start_date) VALUES
(1,'Silver Points','Earn points for every purchase', '2025-01-01'),
(2,'Gold Rewards','Extra benefits for frequent buyers', '2025-02-01'),
(3,'Platinum Club','Exclusive deals for top customers', '2025-03-01'),
(4,'Health Perks','Health-focused reward program', '2025-04-01'),
(5,'Family Care','Rewards for family purchases', '2025-05-01'),
(6,'Wellness Points','Points on wellness products', '2025-06-01'),
(7,'VIP Club','Special privileges for loyal customers', '2025-07-01'),
(8,'Happy Shopper','Discounts and gifts', '2025-08-01'),
(9,'Mega Saver','Bulk purchase rewards', '2025-09-01'),
(10,'Daily Bonus','Points for daily visits', '2025-10-01'),
(11,'Healthy Life','Encouraging healthy lifestyle', '2025-11-01'),
(12,'Smart Shopper','Intelligent shopping rewards', '2025-11-15'),
(13,'Exclusive Deals','Limited-time promotions', '2025-12-01'),
(14,'Loyalty Plus','Extra loyalty benefits', '2025-12-05'),
(15,'Premier Care','Premium rewards program', '2025-12-10');

-- Insert data into branch
INSERT INTO branch (branch_id, branch_name, location, opening_date,
C_company_id) VALUES
(1,'Central','Tashkent', '2024-01-01', 1),
(2,'North','Tashkent', '2024-02-01', 1),
(3,'South','Samarkand', '2024-03-01', 2),
(4,'East','Bukhara', '2024-04-01', 3),
(5,'West','Fergana', '2024-05-01', 4),
(6,'Downtown','Andijan', '2024-06-01', 5),
(7,'City Center','Namangan', '2024-07-01', 6),
(8,'Mall Branch','Nukus', '2024-08-01', 7),
(9,'Airport','Termez', '2024-09-01', 8),
(10,'Station','Karshi', '2024-10-01', 9),
```

```
(11,'University','Jizzakh', '2024-11-01', 10),
(12,'Market','Kokand', '2024-12-01', 11),
(13,'Plaza','Chirchik', '2025-01-01', 12),
(14,'High Street','Navoi', '2025-02-01', 13),
(15,'Riverfront','Margilan', '2025-03-01', 14);

-- Insert data into promotion
INSERT INTO promotion (promo_id, promo_name, start_date, end_date,
promo_type, target_customer, C_company_id) VALUES
(1,'Winter Sale','2025-12-01','2025-12-31','Discount','All',1),
(2,'Health Week','2025-11-01','2025-11-07','Points Bonus','Members',2),
(3,'Buy 1 Get 1','2025-10-01','2025-10-15','Offer','All',3),
(4,'Summer Discount','2025-06-01','2025-06-30','Discount','All',4),
(5,'VIP Rewards','2025-09-01','2025-09-30','Exclusive','VIP',5),
(6,'Flash Sale','2025-12-10','2025-12-12','Discount','All',6),
(7,'Healthy Points','2025-08-01','2025-08-15','Points
Bonus','Members',7),
(8,'Anniversary Offer','2025-07-01','2025-07-07','Discount','All',8),
(9,'New Year Promo','2025-12-20','2025-12-31','Discount','All',9),
(10,'Family Deal','2025-05-01','2025-05-10','Offer','Families',10),
(11,'Student Week','2025-04-01','2025-04-07','Discount','Students',11),
(12,'Platinum Bonus','2025-03-01','2025-03-15','Points
Bonus','Platinum',12),
(13,'Clearance','2025-01-10','2025-01-20','Discount','All',13),
(14,'Summer Fun','2025-06-15','2025-06-25','Offer','All',14),
(15,'Daily Rewards','2025-12-01','2025-12-31','Points Bonus','All',15);

-- Insert data into employee
INSERT INTO employee (employee_id, full_name, position, B_branch_id)
VALUES
(1,'Aliyev Kamol','Manager',1),
(2,'Karimova Lola','Cashier',1),
(3,'Saidov Rustam','Pharmacist',2),
(4,'Nazarova Dildora','Sales Rep',2),
(5,'Olimov Azamat','Manager',3),
(6,'Bekova Sevara','Cashier',3),
(7,'Tursunov Jasur','Pharmacist',4),
(8,'Rashidova Malika','Sales Rep',4),
(9,'Yusupov Anvar','Manager',5),
(10,'Shukurova Nilufar','Cashier',5),
(11,'Islomov Sardor','Pharmacist',6),
(12,'Juraeva Nigora','Sales Rep',6),
(13,'Akbarov Timur','Manager',7),
(14,'Rahimova Laylo','Cashier',7),
(15,'Murodov Bekzod','Pharmacist',8);

-- Insert data into product
INSERT INTO product (product_id, product_name, product_description,
price, PK_category_id, B_branch_id) VALUES
(1,'Vitamin C','Boost immunity',15.50,1,1),
(2,'Pain Reliever','Reduces pain',10.00,2,1),
(3,'Cold Syrup','Treats cold',12.50,3,2),
(4,'Moisturizer','Skin care cream',20.00,4,2),
(5,'Omega 3','Heart health',25.00,5,3),
```

```
(6,'Bandage','First aid',5.00,6,3),
(7,'Probiotics','Digestive health',18.00,7,4),
(8,'Toothpaste','Oral care',4.50,8,4),
(9,'Allergy Relief','Relieves allergy',13.00,9,5),
(10,'Slim Tea','Weight loss',22.00,10,5),
(11,'Eye Drops','Eye care',8.50,11,6),
(12,'Heart Tablets','Heart health',30.00,12,6),
(13,'Kids Vitamins','Children care',14.00,13,7),
(14,'Herbal Tea','Herbal remedy',10.50,14,7),
(15,'Insulin','Diabetes care',50.00,15,8);

-- Insert data into customer
INSERT INTO customer (customer_id, full_name, phone, email, reg_date,
total_points, LP_program_id, E_employee_id) VALUES
(1,'Bekzod Tursunov','998901112233','bekzod@mail.com','2025-01-
01',120,1,1),
(2,'Laylo Karimova','998902223344','laylo@mail.com','2025-01-
05',80,2,2),
(3,'Azamat Olimov','998903334455','azamat@mail.com','2025-02-
01',150,3,3),
(4,'Malika Rashidova','998904445566','malika@mail.com','2025-02-
15',60,4,4),
(5,'Anvar Yusupov','998905556677','anvar@mail.com','2025-03-
01',200,5,5),
(6,'Nilufar Shukurova','998906667788','nilufar@mail.com','2025-03-
15',90,6,6),
(7,'Sardor Islomov','998907778899','sardor@mail.com','2025-04-
01',75,7,7),
(8,'Nigora Juraeva','998908889900','nigora@mail.com','2025-04-
15',50,8,8),
(9,'Timur Akbarov','998909990011','timur@mail.com','2025-05-
01',130,9,9),
(10,'Laylo Rahimova','998910001122','laylo2@mail.com','2025-05-
15',110,10,10),
(11,'Bekzod Murodov','998911112233','bekzod2@mail.com','2025-06-
01',140,11,11),
(12,'Dildora Nazarova','998912223344','dildora@mail.com','2025-06-
15',95,12,12),
(13,'Rustam Saidov','998913334455','rustam@mail.com','2025-07-
01',85,13,13),
(14,'Sevara Bekova','998914445566','sevara@mail.com','2025-07-
15',120,14,14),
(15,'Jasur Tursunov','998915556677','jasur@mail.com','2025-08-
01',100,15,15),
(16, 'Zokirov Dilshod', '998916666666', 'dilshod@mail.com', '2025-09-
01', 0, 1, 1),
(17, 'Usmonova Madina', '998917777777', 'madina@mail.com', '2025-09-
02', 0, 2, 2);

-- Insert data into reward
INSERT INTO reward (reward_id, reward_name, reward_description,
points_required, C_customer_id, LP_program_id) VALUES
(1,'5% Discount','Get 5% off on next purchase',50,1,1),
(2,'Free Delivery','No delivery fee',30,2,2),
```

```
(3,'10% Discount','10% off selected products',100,3,3),
(4,'Health Kit','Free health kit',120,4,4),
(5,'Gift Voucher','$10 gift voucher',80,5,5),
(6,'Wellness Package','Free wellness package',150,6,6),
(7,'Extra Points','Earn 50 bonus points',50,7,7),
(8,'Exclusive Access','VIP event access',200,8,8),
(9,'Free Product','Get a free product',70,9,9),
(10,'Seasonal Gift','Seasonal gift item',90,10,10),
(11,'Loyalty Badge','Special loyalty badge',40,11,11),
(12,'Special Discount','15% discount',110,12,12),
(13,'Free Consultation','Health consultation',60,13,13),
(14,'Premium Pack','Premium gift pack',130,14,14),
(15,'Daily Bonus','Extra points for daily purchase',20,15,15);

-- Insert data into points
INSERT INTO points (points_id, points_status, expiry_date, R_reward_id)
VALUES
(1,'Active','2025-12-31',1),
(2,'Active','2025-12-31',2),
(3,'Active','2025-12-31',3),
(4,'Active','2025-12-31',4),
(5,'Active','2025-12-31',5),
(6,'Active','2025-12-31',6),
(7,'Active','2025-12-31',7),
(8,'Active','2025-12-31',8),
(9,'Active','2025-12-31',9),
(10,'Active','2025-12-31',10),
(11,'Active','2025-12-31',11),
(12,'Active','2025-12-31',12),
(13,'Active','2025-12-31',13),
(14,'Active','2025-12-31',14),
(15,'Active','2025-12-31',15);

-- Insert data into transaction
INSERT INTO transaction (transaction_id, transaction_date,
total_amount, payment_method, points_earned, E_employee_id,
C_customer_id) VALUES
(1,'2025-12-01 10:00:00',120.50,'Cash',10,1,1),
(2,'2025-12-02 11:00:00',75.00,'Card',5,2,2),
(3,'2025-12-03 12:30:00',150.00,'Card',15,3,3),
(4,'2025-12-04 14:00:00',60.00,'Cash',6,4,4),
(5,'2025-12-05 15:15:00',200.00,'Card',20,5,5),
(6,'2025-12-06 16:30:00',90.00,'Cash',9,6,6),
(7,'2025-12-07 17:45:00',75.50,'Card',7,7,7),
(8,'2025-12-08 09:30:00',50.00,'Cash',5,8,8),
(9,'2025-12-09 11:20:00',130.00,'Card',13,9,9),
(10,'2025-12-10 13:40:00',110.00,'Cash',11,10,10),
(11,'2025-12-11 14:50:00',140.00,'Card',14,11,11),
(12,'2025-12-12 15:55:00',95.00,'Cash',9,12,12),
(13,'2025-12-13 10:10:00',85.00,'Card',8,13,13),
(14,'2025-12-14 11:25:00',120.00,'Cash',12,14,14),
(15,'2025-12-15 12:40:00',100.00,'Card',10,15,15);

-- Insert data into shop_at
```

```
INSERT INTO shop_at (C_customer_id, B_branch_id) VALUES
(1,1),(2,1),(3,2),(4,2),(5,3),(6,3),(7,4),(8,4),(9,5),(10,5),
(11,6),(12,6),(13,7),(14,7),(15,8),(1, 2),(1, 3),(2, 3),(2, 4),(3,
1),(3, 4);

-- Insert data into targets
INSERT INTO targets (P_promo_id, C_customer_id) VALUES
(1,1),(2,2),(3,3),(4,4),(5,5),(6,6),(7,7),(8,8),(9,9),(10,10),
(11,11),(12,12),(13,13),(14,14),(15,15);

-- Insert data into get_notify
INSERT INTO get_notify (P_promo_id, C_customer_id) VALUES
(1,1),(2,2),(3,3),(4,4),(5,5),(6,6),(7,7),(8,8),(9,9),(10,10),
(11,11),(12,12),(13,13),(14,14),(15,15),(1, 16),
(2, 17);

-- Insert data into feature_be_feature
INSERT INTO feature_be_feature (P_promo_id, R_reward_id) VALUES
(1,1),(2,2),(3,3),(4,4),(5,5),(6,6),(7,7),(8,8),(9,9),(10,10),
(11,11),(12,12),(13,13),(14,14),(15,15);

-- Insert data into include_items
INSERT INTO include_items (T_transaction_id, P_product_id) VALUES
(1,1),(2,2),(3,3),(4,4),(5,5),(6,6),(7,7),(8,8),(9,9),(10,10),
(11,11),(12,12),(13,13),(14,14),(15,15);

-- Insert data into appear_in
INSERT INTO appear_in (T_transaction_id, P_product_id) VALUES
(1,1),(2,2),(3,3),(4,4),(5,5),(6,6),(7,7),(8,8),(9,9),(10,10),
(11,11),(12,12),(13,13),(14,14),(15,15),(2, 1),(3, 1),(4, 2),(5, 2),(6,
3);

-- Insert data into be_earned_through
INSERT INTO be_earned_through (T_transaction_id, P_points_id) VALUES
(1,1),(2,2),(3,3),(4,4),(5,5),(6,6),(7,7),(8,8),(9,9),(10,10),
(11,11),(12,12),(13,13),(14,14),(15,15);
```

8 SQL Representative Queries: Database Testing

1. Customers with total spending greater than 100 and summarizes their overall purchase value and earned loyalty points

```
SELECT
    c.customer_id,
    c.full_name,
    SUM(t.total_amount) AS total_spent,
    SUM(t.points_earned) AS total_points_earned
FROM customer c
JOIN transaction t ON c.customer_id = t.C_customer_id
```

```
GROUP BY c.customer_id, c.full_name
HAVING SUM(t.total_amount) > 100;
```

2. Top 5 customers by total purchase value

```
SELECT
    c.customer_id,
    c.full_name,
    SUM(t.total_amount) AS total_spent
FROM customer c
JOIN transaction t ON c.customer_id = t.C_customer_id
GROUP BY c.customer_id, c.full_name
ORDER BY total_spent DESC
LIMIT 5;
```

3. Branch performance: number of employees and products per branch

```
SELECT
    b.branch_id,
    b.branch_name,
    COUNT(DISTINCT e.employee_id) AS total_employees,
    COUNT(DISTINCT p.product_id) AS total_products
FROM branch b
LEFT JOIN employee e ON b.branch_id = e.B_branch_id
LEFT JOIN product p ON b.branch_id = p.B_branch_id
GROUP BY b.branch_id, b.branch_name;
```

4. Promotions and how many customers they target

```
SELECT
    pr.promo_name,
    COUNT(t.C_customer_id) AS targeted_customers
FROM promotion pr
LEFT JOIN targets t ON pr.promo_id = t.P_promo_id
GROUP BY pr.promo_name
HAVING COUNT(t.C_customer_id) >= 1;
```

5. Customers who received notifications but did NOT make a transaction

```
SELECT
    c.customer_id,
    c.full_name
FROM customer c
JOIN get_notify gn ON c.customer_id = gn.C_customer_id
WHERE c.customer_id NOT IN (
    SELECT DISTINCT C_customer_id FROM transaction
);
```

6. Products that were sold more than once

```
SELECT
```

```
    p.product_id,
    p.product_name,
    COUNT(ai.T_transaction_id) AS times_sold
FROM product p
JOIN appear_in ai ON p.product_id = ai.P_product_id
GROUP BY p.product_id, p.product_name
HAVING COUNT(ai.T_transaction_id) > 1;
```

7. Loyalty programs with average customer points

```
SELECT
    lp.program_name,
    AVG(c.total_points) AS avg_points
FROM loyalty_program lp
JOIN customer c ON lp.program_id = c.LP_program_id
GROUP BY lp.program_name;
```

8. Employees who handled transactions worth more than average

```
SELECT
    e.employee_id,
    e.full_name,
    SUM(t.total_amount) AS handled_amount
FROM employee e
JOIN transaction t ON e.employee_id = t.E_employee_id
GROUP BY e.employee_id, e.full_name
HAVING SUM(t.total_amount) > (
    SELECT AVG(total_amount) FROM transaction
);
```

9. Rewards linked to promotions and their required points

```
SELECT
    pr.promo_name,
    r.reward_name,
    r.points_required
FROM promotion pr
JOIN feature_be_feature fbf ON pr.promo_id = fbf.P_promo_id
JOIN reward r ON fbf.R_reward_id = r.reward_id
ORDER BY r.points_required DESC;
```

10. Customers who shop at multiple branches

```
SELECT
    c.customer_id,
    c.full_name,
    COUNT(sa.B_branch_id) AS branches_visited
FROM customer c
JOIN shop_at sa ON c.customer_id = sa.C_customer_id
```

```
GROUP BY c.customer_id, c.full_name
HAVING COUNT(sa.B_branch_id) > 1;
```

References

[1] IKEA Family Program [WWW-17.10.2025] - https://www.ikea.com/gb/en/ikea-family/
[2] Starbucks Rewards [WWW-17.10.2025] - https://www.starbucks.com/rewards
[3] Nike Membership [WWW-17.10.2025] - https://www.nike.com/gb/membership
[4] Korzinka Plus [WWW-17.10.2025] - https://korzinka.uz/en/loyalty